k-means Clustering Analysis Report

Dataset: data.npy | Dimensions: 468 × 9 | k range: 1–15

(a) Selection of k

The elbow method selects the value of k at which the marginal reduction in the within-cluster sum of squared errors (SSE) ceases to be substantial. Geometrically, the algorithm draws the chord between the endpoints (k = 1, SSE = 605,274) and (k = 15, SSE = 14,062), then identifies the data point whose perpendicular distance from this chord is maximised. This procedure is applied to the SSE curve in Figure 1.

The curve exhibits a pronounced inflection between k = 5 and k = 6. SSE falls by over 54,000 units when moving from k = 5 (SSE = 71,338) to k = 6 (SSE = 16,710), a relative decrease of ~76 %. By contrast, every subsequent increment yields a decrease of under 400 units, less than 2.5 % of the remaining SSE. The geometric algorithm selects k = 6, consistent with this visual inspection.

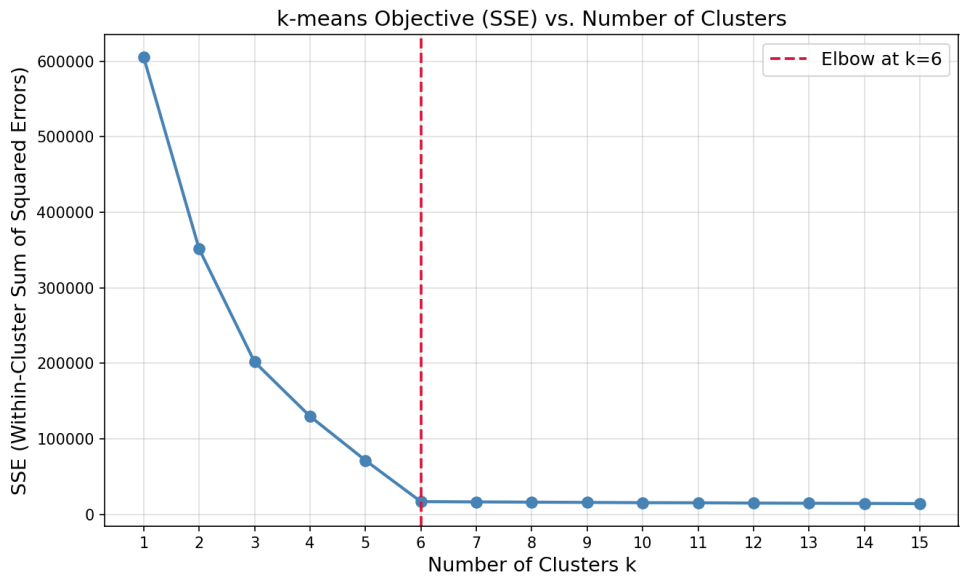

Figure 1. SSE vs. k for k ∈ {1,…,15}. The dashed red line marks the elbow at k = 6, where the curve transitions from steep to flat.

(b) Structural and Statistical Assumptions of k-means

k-means optimises the within-cluster sum of squared Euclidean distances [1]. This objective encodes several assumptions:

- Spherical, isotropic clusters. The squared Euclidean distance treats all dimensions equally, implicitly assuming clusters of similar shape and variance in all directions.
- Equal cluster volumes. The Voronoi partition assigns each point to its nearest centroid; clusters with very different densities or sizes are poorly captured.
- Unimodal, convex structure. Each cluster is assumed to be contiguous and convex, meaning non-convex manifolds or ring-shaped clusters cannot be recovered.
- Fixed, known k. The number of clusters is treated as a hyperparameter rather than inferred from data.

(c) Limitations of the Elbow Method

The elbow procedure has several well-documented weaknesses:

- Scale sensitivity. SSE is not normalised, so the apparent sharpness of the elbow is confounded by data scale and dimensionality. Pre-processing choices (e.g., standardisation) materially alter the curve.
- Ambiguity. Many real-world curves produce no clear elbow; the geometric maximum-distance heuristic then amplifies noise rather than detecting genuine structure.
- No statistical baseline. The method does not compare observed SSE against a null distribution, so it cannot distinguish genuine cluster separation from random fluctuation [2].
- Single-run instability. Although k-means++ [3] with n_init = 10 mitigates local minima, the SSE curve itself can shift across random seeds, making the selected k sensitive to initialisation.

(d) Methodological Improvements and Alternative Validation

Several principled alternatives exist:

- Silhouette coefficient [4]. Measures cohesion and separation jointly, with values in [−1, 1]; the k maximising the average silhouette is preferred, and individual silhouette profiles reveal cluster quality.
- Gap statistic [2]. Compares the log-SSE of the fitted solution against its expectation under a null reference distribution (uniform samples). The selected k is the smallest value for which the gap exceeds the gap at k+1 minus one standard error, providing a formal stopping rule.
- Information-theoretic criteria. BIC or AIC applied to a Gaussian mixture model (GMM) [5] penalise model complexity, enabling direct comparison across k while accounting for parameter count.
- Stability-based selection [6]. Repeat clustering on bootstrap resamples and measure the variation of information or adjusted Rand index between pairs of solutions; genuinely stable k values yield low variation.

(e) When k-means Is Inappropriate: Failure Modes and Detection

k-means is unsuitable under three well-known conditions:

- Non-convex geometry (e.g., two interlocking rings, crescent shapes). Because Voronoi cells are convex, k-means cannot recover manifolds that are not linearly separable [7]. Detection: apply spectral clustering or DBSCAN [8] alongside k-means and compare Normalised Mutual Information (NMI); large discrepancies flag geometric inadequacy.
- Heterogeneous covariance structure. When clusters differ in orientation or elongation, the spherical assumption causes misassignment at cluster boundaries. Detection: fit a full GMM with tied, diagonal, and full covariance and compare BIC [5]; preference for full covariance indicates anisotropy.
- Varying density and noise. DBSCAN's classic 'two dense blobs with a bridge of noise' example is irrecoverable by k-means, which greedily absorbs outliers into clusters. Detection: inspect the k-nearest-neighbour distance distribution [8]; heavy-tailed distributions suggest heterogeneous density and motivate density-based methods.
- High-dimensional spaces. In high dimensions, Euclidean distance concentrates [9], so all pairwise distances become approximately equal and cluster boundaries dissolve. Detection: measure the relative contrast (max dist – min dist) / min dist across a random subsample; values near zero indicate distance concentration and call for dimensionality reduction (PCA, UMAP) prior to clustering.

In the present dataset (468 × 9, k = 6 with SSE dropping 76 % at the elbow), the sharp transition in SSE and the moderate dimensionality suggest k-means is a defensible choice. Nonetheless, silhouette

analysis and GMM comparison are recommended to confirm cluster validity before substantive interpretation.

References

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